

Democratizing AI Explainability through a Hybrid XAI-LLM Framework for Accessible and Reliable Explanations

Louenas Bounia¹[0009–0006–8771–0401] and Mor Ndour¹

LIPN, UMR CNRS 7030, Université Sorbonne Paris Nord, Villetaneuse, France
name@lipn.univ-paris13.fr

Abstract. This work proposes an approach to democratize the explainability of AI models using large language models (LLMs), with an initial application to decision trees. Traditional explainability techniques, often technical, remain largely inaccessible to non-experts. By combining LLMs with XAI methods such as SHAP, we generate simple, natural-language explanations that are understandable to a non-specialist audience. Our experimental protocol applies this approach to decision trees to produce contextualized and personalized explanations. A dual reliability assessment, computed through both qualitative evaluation and formal verification of the consistency between the explanations and the model’s internal reasoning, is provided to assess the trustworthiness of this democratization. The results show an improvement in the accessibility and relevance of explanations for non-experts, strengthening the transparency of AI models.

Keywords: Explainable AI (XAI) · Large Language Models (LLMs) · AI Democratization · Human-Computer Interaction (HCI).

1 Introduction

Artificial intelligence (AI) systems are now widely used across sectors such as healthcare, finance, education, transportation, and public services. By 2025, most global organizations utilize AI in at least one business function, and generative AI is being adopted particularly rapidly. These developments are transforming traditional practices into more efficient and autonomous solutions.

As AI systems become increasingly prevalent, it is essential that their decisions are both accurate and interpretable by humans. This requirement has driven the development of Explainable Artificial Intelligence (XAI), which seeks to make algorithmic reasoning transparent and trustworthy (29; 22). Clear explanations are crucial for establishing user trust, ensuring accountability, and supporting effective human–AI collaboration.

In recent years, XAI has grown rapidly, attracting significant attention in both research and industry (23). Existing approaches aim to improve model

transparency; examples include model-agnostic methods such as LIME (26), SHAP (20), and Anchors (27). Although widely adopted, these methods have notable limitations: their outputs may lack reproducibility and fidelity guarantees, raising concerns about their scientific robustness and intrinsic interpretability (28; 21).

In contrast, formal XAI approaches focus on generating explanations by analyzing the internal structure of models, providing rigorous and consistent insights into their operation (31; 3; 11; 13; 7). These methods offer desirable theoretical properties and practical algorithms for computing explanations. However, they also face important limitations: they may not scale efficiently and can struggle to produce explanations for complex AI models in a reasonable time, particularly for tree-based models such as random forests (5) or boosted trees, as well as deep learning models including convolutional or Bayesian neural networks.

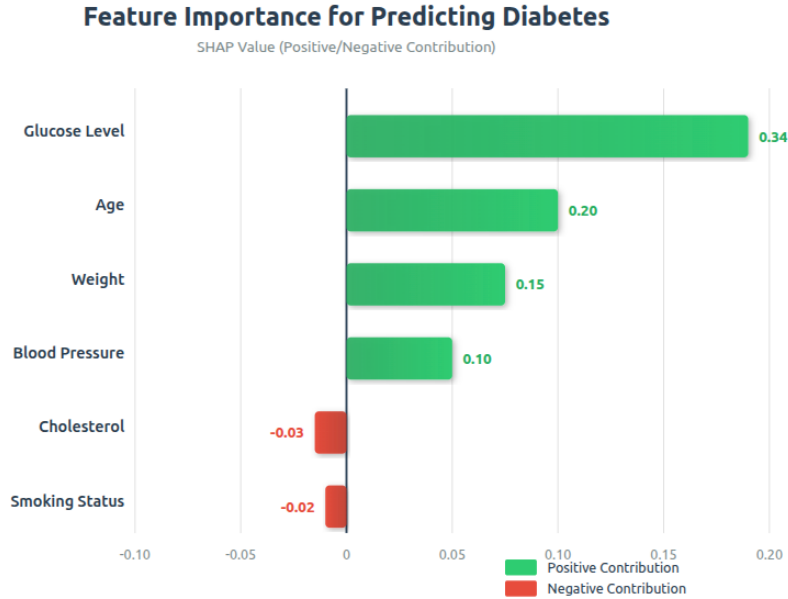


Fig. 1: SHAP feature importance visualization for diabetes prediction

In summary, the common objective of the different Explainable Artificial Intelligence (XAI) methods is to open the “black box” of machine learning models by providing understandable explanations of their decisions. However, despite their methodological differences, these approaches share a significant shortcoming: the explanations they generate often remain too technical and difficult to access for non-expert users (18; 9).

This lack of accessibility constitutes a major barrier to the adoption of explainable AI by non-expert users. For example, Figure 1 illustrates a SHAP explanation for a model h predicting the risk of diabetes: although informative from an algorithmic perspective, it remains inaccessible without technical expertise (24). This interpretability gap hinders understanding (8), limits trust, and slows the democratization of artificial intelligence explainability (15; 14).

However, to be truly useful, explanations must be accessible and democratized for the end user. Explainability cannot be limited to a technical approach intended for experts: it must be part of a social process where explanations are adapted to the level of understanding (17), the needs, and the context of each user (1). This requires transforming the raw outputs of XAI methods into understandable, personalized, and actionable content, thus enabling all users, whether experts or not, to appropriate and benefit from the explanations provided by AI systems.

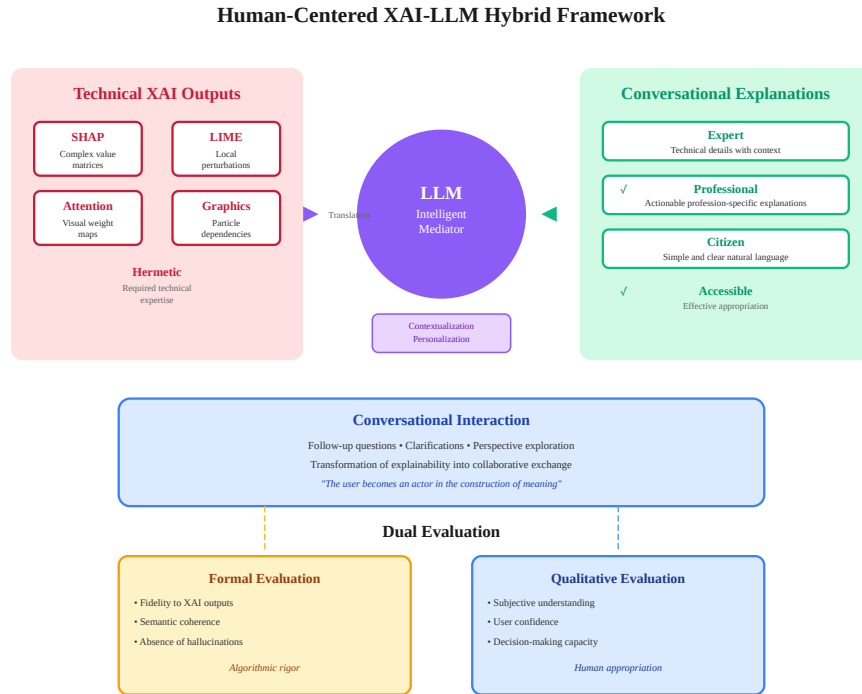


Fig. 2: Human-Centered XAI-LLM Hybrid Framework Architecture

To address this gap, we propose a human-centered hybrid XAI-LLM framework (Figure 2). Our approach leverages the communicative abilities of large

language models to convert technical XAI outputs into accessible, personalized, and conversational explanations (Figure 2). Traditional XAI techniques—such as SHAP, LIME, attention maps, and formal methods—often produce outputs that are difficult for non-experts to interpret, including complex value matrices, partial dependence plots, or visual weight maps. By acting as intelligent mediators, LLMs translate these technical representations into natural language narratives tailored to each user’s needs, whether for technical experts, business professionals seeking actionable insights, or non-expert end users requiring simple and clear explanations.

Beyond technical interpretability, our approach prioritizes user experience and cognitive accessibility through contextual personalization. We recognize that true explainability is not measured solely by the mathematical fidelity of the outputs, but by the effective appropriation by users in their usage context (8; 22). As shown in Figure 2, our framework integrates conversational interaction mechanisms allowing follow-up questions, targeted clarifications, or exploration of different explanatory perspectives, transforming explainability from a unidirectional process into a true collaborative exchange where “the user becomes an active participant in the construction of meaning.”

Finally, our methodology aims to simultaneously ensure the human comprehensibility of explanations, while also quantifying the risk associated with generating these explanations through a dual evaluation combining formal analyses and qualitative assessments involving end users (Figure 2). The formal evaluation measures the fidelity of LLM explanations to the reasoning of the learning model, assessing semantic coherence by limiting hallucinations to ensure algorithmic rigor. In parallel, the qualitative evaluation mobilizes user studies to measure subjective understanding, the induced level of trust, and effective decision-making capability, thus ensuring human appropriation of the received explanations.

Contributions. Our contribution consists of a human-centered XAI-LLM hybrid framework enabling the democratization of explanations generated by an AI model, ensuring that they are both understandable and reliable. Our architecture (Figure 3) consists of three interdependent modules: a **Generative Explanation System** that uses an LLM to transform technical XAI outputs into conversational explanations (32), a **Parser Module** that extracts structured logical rules via few-shot learning (6) while filtering potential hallucinations, and a **Dual Evaluation System** combining a qualitative analysis of semantic coherence (19) with formal verification via $1 - \epsilon_{h,x}$ (4) on Boolean circuits (30). For a first application of explanation democratization, we applied the process described in Figure 3 to decision tree models. We evaluated our hybrid framework on 7 datasets (COMPAS, Placement, Heart, Glass, Diabetes, Tic-Tac-Toe, Vote House 84) using five LLMs (Gemini-1.5-Flash, GPT-4o, Deepseek-Chat, Claude-Sonnet-4, Grok-3). The results show that our approach achieves on average $> 93\%$ in qualitative evaluation and 68% in formal fidelity via the metric

$1 - \epsilon_{h,x}$. These results establish that our hybrid framework reconciles cognitive accessibility and algorithmic transparency, paving the way toward truly democratized explainability.

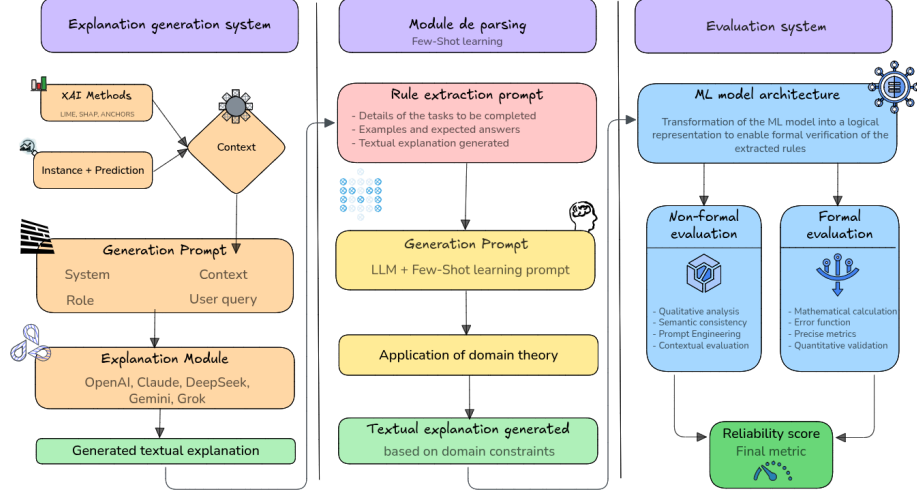


Fig. 3: Pipeline for democratized explainability with generation, parsing, and evaluation modules

2 Methodology

Democratizing explanation with LLMs. From a human-computer interaction perspective, LLMs can support the creation of adaptive explanatory interfaces. For instance 1, an LLM can translate SHAP outputs into natural language, such as: ‘*The patient presents a risk of diabetes mainly due to high glucose levels, age, and weight. Their cholesterol level and non-smoking status slightly reduce this probability.*’ This conversational explanation makes abstract XAI outputs more understandable, enhancing cognitive accessibility for end users.

However, uncontrolled LLM outputs may reduce transparency: they can produce hallucinated explanations (34; 12) or prioritize fluency over fidelity to the underlying model h . Existing approaches rarely demonstrate their effectiveness in user-centered evaluations (10). These challenges highlight the need for rigorously designed hybrid interfaces that ensure both accessibility and reliability. In our framework, systematic formal validation is incorporated to maintain consistency between conversational explanations and the model’s internal structure.

3 Pipeline for democratized explainability: LLM generation and evaluation systems

We present a human-centered hybrid XAI-LLM framework designed to make AI model explanations both accessible and reliable. To ensure alignment with the model’s reasoning, the framework incorporates formal verification. Its architecture (Figure 3) consists of three interdependent modules, which generate user-friendly explanations while systematically evaluating their fidelity.

- **Generative Explanation System:** The first module receives as input the model architecture h for a given instance x , the corresponding XAI output (e.g., SHAP feature importance), and the model prediction $h(x)$. A structured prompt specifies the system’s role, context, and the user query (33). Using an LLM (OpenAI, Claude, DeepSeek, Gemini, Grok), this module generates a natural language explanation (32), converting technical XAI outputs into accessible and user-friendly narratives.
- **Parsing Module:** This module mediates between accessibility and formal rigor. A rule extraction prompt defines the task, provides annotated examples, and submits the LLM-generated explanation. Using few-shot learning (6), the LLM extracts logical rules from the explanation. These rules are then structured according to domain constraints, ensuring they comply with the domain on which the classifier was trained (see (2)). Out-of-domain rules are filtered, and the rules are converted into a compatible binary representation. This process reduces potential hallucinations by retaining only domain-consistent rules.
- **Evaluation System:** This module provides dual validation of explanation reliability. First, the model architecture is transformed into a Boolean circuit using a CNF representation (30), enabling two complementary evaluations. The *non-formal evaluation* employs a separate LLM to assess the semantic coherence of the extracted rules qualitatively (19). The *formal evaluation* computes an explanation error function $1 - \epsilon_{h,x}$ for a model h (see (4)), instance x , and rule set S . This metric measures how well the extracted rules align with the model’s internal operations.

4 Applications to Decision Trees

As a first application of our architecture, we chose the machine learning model of decision trees. The choice of decision trees as the first model is supported by the ease of performing formal verification via linear-time evaluation of the function $\epsilon_{h,x}$ (4; 25), and the possibility of also transforming the decision tree into a CNF formula in linear time (30), as well as the ease of extracting the domain on which the tree is based (its theory domain) as illustrated in (2). Decision trees represent an ideal ground for a first application of our approach.

4.1 Experimental Protocol

To evaluate the effectiveness and reliability of our hybrid framework, we conducted experiments on 7 benchmark datasets: COMPAS, Placement, Heart, Glass, Diabetes, Tic-Tac-Toe, and Vote House 84. For each dataset, a decision tree was trained via scikit-learn (DecisionTreeClassifier) as the target model h . A Grid Search strategy was used to optimize the hyperparameters and select the most performant model for each dataset.

For each trained model, we selected 15 representative instances and computed the classic local XAI explanations (SHAP, LIME, Anchors). The architecture of each decision tree was transformed into a CNF representation (16) via SAT encoding, a crucial step to enable subsequent formal verification. This transformation provides the LLM with a structured representation of the model’s internal architecture, thus guiding the generation of explanations faithful to the actual behavior of h and limiting potential hallucinations. Orthogonal DNFs representing the decision tree and its negation were also computed for formal evaluation via the function $\epsilon_{h,x}$.

We tested five different LLMs (Gemini-1.5-Flash, GPT-4o, Deepseek-Chat, Claude-Sonnet-4, Grok-3) to generate conversational explanations and extract logical rules via few-shot learning. To ensure neutrality and avoid potential biases from the conversational interfaces of these LLMs, we used their official APIs (which we paid for) via direct queries. For each instance and each LLM, we applied our full pipeline: generation of textual explanation by providing the CNF model architecture to the LLM, extraction of rules via the parsing module, application of domain theory to filter out-of-domain rules (thus limiting hallucinations by ensuring that only rules respecting the learning domain constraints are retained), followed by dual evaluation.

The dual evaluation constitutes an essential control mechanism to ensure that the generated explanations do not mislead users. The non-formal evaluation is carried out using a second LLM that checks the logical coherence of each extracted rule with the model architecture in CNF. The non-formal reliability score corresponds to the ratio between the number of coherent rules and the total number of extracted rules. The formal evaluation, on the other hand, provides mathematically rigorous validation calculated via the explanation error function $\epsilon_{h,x}(S)$ using the extracted rule set S , giving the score $(1 - \epsilon_{h,x}) \times 100$, which precisely quantifies the coherence between the extracted rules and the model’s internal architecture. This dual validation allows detecting and quantifying potential divergences between the explanation provided to the user and the actual behavior of the model, thereby protecting the user from misleading explanations that could induce erroneous decisions.

The experiments were conducted using Python code executed on a machine equipped with an Intel(R) Core i9-9900 processor running at 3.1 GHz and 64 GB of RAM.

4.2 Results

Table 1 shows the average reliability scores (in %) of five LLMs across seven datasets. Each cell reports a pair of scores: the *non-formal score*, representing the qualitative coherence of the extracted rules (evaluated by a separate LLM), and the *formal score*, measuring fidelity to the model’s internal architecture using the metric $1 - \epsilon_{h,x}$. Each score is averaged over 15 instances per dataset. **Green** cells indicate the highest formal scores, while **yellow** cells indicate the highest non-formal scores. This presentation highlights both perceived coherence and formal fidelity, helping to interpret the reliability of explanations for end users.

- **Columns:** The first column indicates the name of the tested dataset. The following five columns correspond respectively to the five evaluated LLMs: Gemini-1.5-Flash, GPT-4o, Deepseek-Chat, Claude-Sonnet-4, and Grok-3.
- **Rows:** The first seven rows correspond to the seven tested datasets (Vote House 84, Diabetes, Tic-Tac-Toe, Heart, Glass, Compas, and Placement). The last row presents the global average calculated across all datasets for each LLM.
- **Cell contents:** Each cell contains two scores expressed as percentages, presented in the format "Non-Formal Score / Formal Score." The non-formal score (first number) measures the qualitative coherence of the extracted rules, verified by a separate LLM. The formal score (second number) quantifies the mathematical fidelity of the rules to the internal architecture of the model h , calculated via the function $(1 - \epsilon_{h,x})$. Each score represents the average computed over 15 instances of the corresponding dataset.

Results Analysis. Table 1 reveals several important trends regarding the reliability of explanations generated by LLMs.

- **Gap between formal vs. non-formal evaluation.** We observe a consistent gap between non-formal scores (83 – 100%) and formal scores (52 – 84%), averaging approximately 30 percentage points. The non-formal evaluation reflects the perceived logical coherence of the extracted rules, whereas the formal evaluation using $\epsilon_{h,x}$ measures the actual alignment with the model’s internal structure. Lower formal scores indicate that, while LLM-generated rules appear coherent, they may partially diverge from the model’s true behavior. These findings highlight the importance of combining user-centered and formal evaluations when assessing explanation reliability. This observation empirically validates the necessity of dual evaluation: relying solely on non-formal evaluation would lead to a dangerously overestimated reliability, potentially misleading the end user. The gap is particularly pronounced on certain datasets such as Heart (100% vs. 61.64% for GPT-4o) or Compas (100% vs. 59.62% for Grok-3), highlighting the risk of an illusion of fidelity.

LLM	Non-Formal Score	Formal Score
<i>Vote House 84</i>		
Gemini-1.5	91.48	62.49
GPT-4o	91.47	66.69
Deepseek	96.22	70.74
Claude-S4	96.33	83.99
Grok-3	99.04	65.50
<i>Diabetes</i>		
Gemini-1.5	88.92	52.84
GPT-4o	97.33	52.99
Deepseek	92.92	52.81
Claude-S4	84.46	53.88
Grok-3	97.33	53.97
<i>Tic-Tac-Toe</i>		
Gemini-1.5	96.17	71.40
GPT-4o	98.93	70.34
Deepseek	97.33	74.82
Claude-S4	99.33	75.48
Grok-3	99.33	73.66
<i>Heart</i>		
Gemini-1.5	87.44	58.52
GPT-4o	100.00	61.64
Deepseek	85.94	59.43
Claude-S4	84.98	63.97
Grok-3	90.11	61.76
<i>Glass</i>		
Gemini-1.5	96.67	57.84
GPT-4o	89.42	58.50
Deepseek	89.83	59.67
Claude-S4	90.67	74.95
Grok-3	98.67	58.75
<i>Compas</i>		
Gemini-1.5	83.53	72.49
GPT-4o	93.11	62.27
Deepseek	77.42	61.87
Claude-S4	86.90	56.82
Grok-3	100.00	59.62
<i>Placement</i>		
Gemini-1.5	91.20	66.82
GPT-4o	83.87	61.08
Deepseek	91.88	65.10
Claude-S4	85.98	56.54
Grok-3	90.27	53.82
OVERALL AVERAGES		
Gemini-1.5	90.77	63.20
GPT-4o	93.45	61.93
Deepseek	90.22	63.49
Claude-S4	89.81	66.52
Grok-3	96.39	60.73

Table 1: Average non-formal and formal reliability scores (in %) for five LLMs across seven datasets (15 instances per dataset).

- **Variability across datasets.** Formal performance varies significantly depending on the dataset: Tic-Tac-Toe and Vote House 84 reach 70 – 84%, whereas Diabetes caps at 52 – 54% for all LLMs. This variability suggests that the structural complexity of the model (tree depth, number of branches) and the nature of the domain (relationships between features) strongly influence the ability of LLMs to produce faithful rules.
- **Comparison of LLMs.** Among the LLMs, Claude-Sonnet-4 achieves the highest formal average (66.52%), reaching 83.99% on the Vote House 84 dataset, indicating strong fidelity to the model. Grok-3 performs best in non-formal evaluation (96.39%) but shows a lower formal score (60.73%), highlighting a gap between perceived and actual reliability. GPT-4o, Gemini-1.5, and Deepseek display balanced performance, with formal scores between 61 – 63% and non-formal scores between 90 – 93%. These results suggest that LLM choice can influence the trustworthiness of explanations depending on whether fidelity or perceived coherence is prioritized.
- **Special cases.** The score of 83.99% (Claude-S4 on Vote House 84) represents the best absolute result, proving that very high fidelity is achievable. Conversely, the universal cap at 53% on Diabetes suggests intrinsic characteristics (possibly complex feature interactions or a particularly deep tree structure) making the generation of faithful rules difficult.

In summary, the results confirm the effectiveness of the framework with average formal scores of 60 – 67%, while highlighting the critical importance of formal evaluation to ensure reliability. The choice of LLM depends on the context: Claude-Sonnet-4 for critical applications requiring high formal fidelity, Grok-3 to prioritize the qualitative coherence of explanations.

4.3 Discussion

The results demonstrate that conversational explanations generated by LLMs can achieve high levels of perceived coherence, making them accessible and understandable for non-expert users. However, the systematic gap between non-formal and formal evaluations reveals a critical risk: explanations that appear convincing may not faithfully reflect the underlying model’s behavior. From an HCI perspective, this gap corresponds to a potential illusion of interpretability, where users may develop unwarranted trust in explanations that are rhetorically coherent but technically inaccurate.

By explicitly combining formal fidelity metrics with user-centered qualitative evaluation, our framework addresses this challenge and provides a more reliable foundation for human–AI interaction. This dual evaluation paradigm supports informed decision-making and encourages calibrated trust rather than blind reliance on AI-generated explanations.

5 Conclusion and Future Work

Conclusion. This paper introduces a hybrid XAI-LLM framework aimed at democratizing AI explainability by bridging formal model explanations and conversational interaction. By combining traditional XAI outputs with the communicative abilities of large language models, our approach transforms technical explanations into accessible, user-friendly narratives while explicitly quantifying their reliability.

Experiments conducted on seven datasets with five LLMs demonstrate that the framework achieves high perceived coherence (above 90%) while maintaining moderate to high formal fidelity (60 – 67%), with peaks reaching 83.99%. The consistent gap between formal and non-formal evaluations empirically confirms the necessity of dual validation to prevent misleading interpretations. Overall, our results show that explainability can be both cognitively accessible and algorithmically grounded, fostering trustworthy human-AI interaction.

Future Work. Our future research is organized around three complementary directions:

- **Enhanced formal verification.** We are currently integrating advanced formal verification mechanisms to maximize fidelity scores. This improvement will be coupled with a dialogic approach, transforming explainability into a bidirectional conversational interaction where users can explore different explanatory perspectives.
- **Extension to ensemble models.** We are extending our democratization approach to more complex architectures (Random Forests, XGBoost, etc.) to generalize our pipeline beyond simple decision trees, paving the way for broader applicability across different machine learning algorithms.
- **Advanced personalization via RAG.** The integration of RAG (Retrieval-Augmented Generation) will enable improved personalization through indexing of representative queries from diverse user profiles. The system will perform similarity search to generate tailored explanations, significantly enhancing the current approach that relies solely on the LLM’s internal capabilities.

Bibliography

- [1] Ali, S., Abuhmed, T., El-Sappagh, S., Muhammad, K., Alonso-Moral, J.M., Confalonieri, R., Guidotti, R., Del Ser, J., Díaz-Rodríguez, N., Herrera, F.: Explainable Artificial Intelligence (XAI): What we know and what is left to attain Trustworthy Artificial Intelligence. *Information Fusion* **99**, 101805 (2023)
- [2] Audemard, G., Lagniez, J.-M., Marquis, P., Szczepanski, N.: Deriving Provably Correct Explanations for Decision Trees: The Impact of Domain Theories. In: *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence (IJCAI-24)*. pp. 3688–3696 (2024)
- [3] Audemard, G., Bellart, S., Bounia, L., Koriche, F., Lagniez, J.-M., Marquis, P.: On the Computational Intelligibility of Boolean Classifiers. In: *Proc. of KR’21*. pp. 74–86 (2021)
- [4] Bounia, L., Koriche, F.: Approximating probabilistic explanations via supermodular minimization. In: *Uncertainty in Artificial Intelligence*. pp. 216–225. PMLR (2023)
- [5] Bounia, L.: Using Submodular Optimization to Approximate Minimum-Size Abductive Path Explanations for Tree-Based Models. In: *The 41st Conference on Uncertainty in Artificial Intelligence (UAI 2025)*. (2025)
- [6] Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J.D., Dhariwal, P., Amodei, D., et al.: Language models are few-shot learners. In: *Advances in Neural Information Processing Systems*. vol. 33, pp. 1877–1901 (2020)
- [7] Darwiche, A.: Logic for Explainable AI. In: *Proceedings of the 38th Annual ACM/IEEE Symposium on Logic in Computer Science (LICS)*. pp. 1–11. IEEE (2023)
- [8] Doshi-Velez, F., Kim, B.: Towards a Rigorous Science of Interpretable Machine Learning. *arXiv preprint arXiv:1702.08608* (2017)
- [9] Ehsan, U., Riedl, M.O.: Human-Centered Explainable AI: Towards a Reflective Sociotechnical Approach. In: *HCI International 2020 - Late Breaking Papers: Multimodality and Intelligence*. pp. 449–466. Springer (2020)
- [10] Gat, Y.O., Calderon, N., Feder, A., Chapanin, A., Sharma, A., Reichart, R.: Faithful Explanations of Black-box NLP Models Using LLM-generated Counterfactuals. In: *The Twelfth International Conference on Learning Representations*. (2024)
- [11] Ignatiev, A., Narodytska, N., Marques-Silva, J.: Abduction-Based Explanations for Machine Learning Models. In: *Proc. of AAAI’19*. pp. 1511–1519 (2019)
- [12] Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Ishii, E., Bang, Y.J., Madotto, A., Fung, P.: Survey of Hallucination in Natural Language Generation. *ACM Computing Surveys* **55**(12), 1–38 (2023)
- [13] Marques-Silva, J.: Logic-Based Explainability in Machine Learning. *ArXiv abs/2211.00541* (2023)

- [14] Kaur, H., Williams, A.C., Thompson, A.L., Lasecki, W.S., Iqbal, S.T., Teevan, J.: “Help Me Help the AI”: Understanding How Explainability Can Support Human-AI Interaction. In: *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. pp. 1–17 (2023)
- [15] Khosravi, H., Buckingham Shum, S., Chen, G., Conati, C., Tsai, Y.-S., Kay, J., Knight, S., Martinez-Maldonado, R., Sadiq, S., Gašević, D.: Explainable Artificial Intelligence in Education. *Computers and Education: Artificial Intelligence* **3**, 100074 (2022)
- [16] Audemard, G., Bellart, S., Bounia, L., Koriche, F., Lagniez, J.-M., Marquis, P.: On the explanatory power of Boolean decision trees. *Data & Knowledge Engineering* **142**, 102088 (2022)
- [17] Kim, M., Kim, S., Kim, J., Song, T.-J., Kim, Y.: Do stakeholder needs differ? - Designing stakeholder-tailored Explainable Artificial Intelligence (XAI) interfaces. *International Journal of Human-Computer Studies* **181**, 103160 (2024)
- [18] Liao, Q.V., Varshney, K.R.: Human-Centered Explainable AI (XAI): From Algorithms to User Experiences. arXiv preprint arXiv:2110.10790 (2021)
- [19] Liu, Y., Iter, D., Xu, Y., Wang, S., Xu, R., Zhu, C.: G-Eval: NLG evaluation using GPT-4 with better human alignment. In: *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. pp. 2511–2522 (2023)
- [20] Lundberg, S.M., Lee, S.-In: A Unified Approach to Interpreting Model Predictions. In: *Advances in Neural Information Processing Systems*. vol. 30, pp. 4765–4774 (2017)
- [21] Marques-Silva, J.: Disproving XAI myths with formal methods. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. vol. 37, pp. 15402–15410 (2023)
- [22] Miller, T.: Explanation in artificial intelligence: Insights from the social sciences. *Artificial Intelligence* **267**, 1–38 (2019)
- [23] Molnar, C.: *Interpretable Machine Learning: A Guide for Making Black Box Models Explainable*. 3rd edn. Leanpub (2025)
- [24] Ridley, M.: Human-Centered Explainable Artificial Intelligence: An ARIST Paper. *Journal of the Association for Information Science and Technology* **75**(8), 927–952 (2024)
- [25] Bounia, L., Setitra, I.: Enhancing the intelligibility of decision trees with concise and reliable probabilistic explanations. *Data & Knowledge Engineering* **156**, 102394 (2025)
- [26] Ribeiro, M.T., Singh, S., Guestrin, C.: “Why Should I Trust You?”: Explaining the Predictions of Any Classifier. In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. pp. 1135–1144 (2016)
- [27] Ribeiro, M.T., Singh, S., Guestrin, C.: Anchors: High-Precision Model-Agnostic Explanations. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. vol. 32, pp. 1527–1535 (2018)
- [28] Rudin, C., Chen, C., Chen, Z., Huang, H., Semenova, L., Zhong, C.: *Interpretable Machine Learning: Fundamental Principles and 10 Grand Challenges*. *Statistics Surveys* **16**, 1–85 (2022). <https://doi.org/10.1214/21-SS133>

- [29] Schmidt, P., Biessmann, F., Teubner, T.: Transparency and Trust in Artificial Intelligence Systems. *Journal of Decision Systems* **29**(4), 260–278 (2020)
- [30] Shi, W., Shih, A., Darwiche, A., Choi, A.: Compiling neural network classifiers into Boolean circuits. In: *Proceedings of the 16th International Conference on Principles of Knowledge Representation and Reasoning*. pp. 565–574 (2019)
- [31] Shih, A., Choi, A., Darwiche, A.: A Symbolic Approach to Explaining Bayesian Network Classifiers. In: *Proc. of IJCAI'18*. (2018)
- [32] Slack, D., Krishna, S., Lakkaraju, H., Singh, S.: Rethinking explainability as a dialogue: Practitioner perspectives of LLM-based conversational interpretability systems. In: *Proceedings of the CHI Conference on Human Factors in Computing Systems*. pp. 1–16. ACM (2023)
- [33] White, J., Fu, Q., Hays, S., Sandborn, M., Olea, C., Gilbert, H., Elnashar, A., Spencer-Smith, J., Schmidt, D.C.: A prompt pattern catalog to enhance prompt engineering with ChatGPT. *arXiv preprint arXiv:2302.11382* (2023)
- [34] Xu, Z., Jain, S., Kankanhalli, M.: Hallucination is Inevitable: An Innate Limitation of Large Language Models. *arXiv preprint arXiv:2401.11817* (2025)